

To see that 1) $\sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = 1$, note that the sum of weights is equal to

$$\sum_{t=0}^T \sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = \sum_{t=0}^T \sum_{\ell \in g} \sum_e \mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}] \quad (55)$$

$$= \mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \sum_{\ell \in g} D_{i,t}^\ell \cdot \sum_e \mathbf{1}\{E_i = e\}] \quad (56)$$

$$= \mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{1}\{t - E_i \in g\}]. \quad (57)$$

It is thus the population regression coefficient on $\mathbf{1}\{t - E_i \in g\}$ from regressing $\mathbf{1}\{t - E_i \in g\}$ on $\mathbf{D}_{i,t}$ and the unit and time fixed effects, which is just one. Similarly, for each $g' \neq g$, the sum of weights is the population regression coefficient on $\mathbf{1}\{t - E_i \in g'\}$ from regressing $\mathbf{1}\{t - E_i \in g\}$ on $\mathbf{D}_{i,t}$ and the unit and time fixed effects, which is zero. To see that the weights from all excluded relative time add up to negative one across cohorts, note that the sum of these weights is equal to

$$\sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g = \mathbf{e}_g^\top \sum_{\ell \in g^{excl}} \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell] \quad (58)$$

$$= \mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E\left[\ddot{\mathbf{D}}_{i,t} \sum_{\ell \in g^{excl}} D_{i,t}^\ell\right] \quad (59)$$

$$= \mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} (1 - \sum_{g \in \mathcal{G}} \sum_{\ell \in g} D_{i,t}^\ell)] \quad (60)$$

$$= -\mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \sum_{g \in \mathcal{G}} \sum_{\ell \in g} D_{i,t}^\ell] \quad (61)$$

$$= -\mathbf{e}_g^\top \sum_{g \in \mathcal{G}} \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{1}\{t - E_i \in g\}] = -1 \quad (62)$$

where the third equality follows from $\sum_{\ell \in g^{excl}} D_{i,t}^\ell + \sum_{g \in \mathcal{G}} \sum_{\ell \in g} D_{i,t}^\ell = \sum_{-T \leq \ell \leq T} D_{i,t}^\ell = 1$. The fourth equality follows from $\sum_t E[\ddot{\mathbf{D}}_{i,t}] = 0$ due to the zero mean property of double-demeaning proved in Lemma 1.

The last line simplifies to negative one because each term in the summation

$$\left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{1}\{t - E_i \in g\}] \quad (63)$$

equals the population regression coefficient on $\mathbf{D}_{i,t}$ from regressing $\mathbf{1}\{t - E_i \in g\}$ on $\mathbf{D}_{i,t}$ and the unit and time fixed effects, which is a column vector with one in the entry corresponding to $\mathbf{1}\{t - E_i \in g\}$. Summing over g , the above expression is equal to a column vector of ones.

Finally, we note that the weight for the never-treated cohort is always zero $\omega_{\infty,\ell}^g = 0$. This is because $D_{i,t}^\ell = 0$ for all ℓ when $E_i = \infty$. $\square$

Proof of Proposition 2

Proof. Under the parallel trends assumption, we can replace each term in Proposition 1 with

$$E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] = CATT_{e,\ell} + \underbrace{E[Y_{i,t}^\infty - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty]}_{=0} \quad (64)$$

for $t = e + \ell$. $\square$

Proof of Proposition 3

Proof. Under the no anticipation assumption, setting the corresponding terms to zero in the linear combination underlying μ_g displayed in Proposition 2. $\square$

Proof of Proposition 4

Proof. Under the assumption of homogeneous treatment effect, we have $CATT_{e,\ell} = ATT_\ell$ for all e . This simplifies Proposition 2 to

$$\sum_{\ell \in g} \left(\sum_e \omega_{e,\ell}^g \right) ATT_\ell + \sum_{g' \neq g} \sum_{\ell \in g'} \left(\sum_e \omega_{e,\ell}^g \right) ATT_\ell + \sum_{\ell \in g^{excl}} \left(\sum_e \omega_{e,\ell}^g \right) ATT_\ell. \quad (65)$$

Recall the weight $\omega_{e,\ell}^g$ is equal to the population regression coefficient on $\mathbf{1}\{t - E_i \in g\}$ from regressing $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ on all the bin indicators, i.e. $\{\mathbf{1}\{t - E_i \in g\}\}_{g \in \mathcal{G}}$ and two-way fixed effects included in (4). Summing over e would imply the weight $\omega_\ell^g := \sum_e \omega_{e,\ell}^g$ is equal to the population regression coefficient on $\mathbf{1}\{t - E_i \in g\}$ from regressing $D_{i,t}^\ell$ on all the bin indicators, i.e. $\{\mathbf{1}\{t - E_i \in g\}\}_{g \in \mathcal{G}}$ and two-way fixed effects included in (4). $\square$

C Supplementary results related to IW estimator

In this section, we first provide detailed definition for the IW estimator, and then state its asymptotic behavior. We then prove its asymptotic normality and derive its asymptotic variance in Section C.1.

Definition 4. The IW estimator for ν_g , a weighted average of $CATT_{e,\ell \in g}$ is constructed via the following three steps. We focus on the setting without never-treated units, using pre-period $s = -1$ and the last-treated cohort as the control cohort $C = \{\max\{E_i\}\}$. The setting with never-treated units can be defined similarly by modifying the interacted specification.

Step 1 Estimate the $CATT_{e,\ell}$ using the interacted specification.

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{e \neq \max\{E_i\}} \sum_{\ell \neq -1} \delta_{e,\ell} (\mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell) + \epsilon_{i,t} \quad (66)$$

$$= \alpha_i + \lambda_t + \mathbf{B}_{i,t}^\top \boldsymbol{\delta} + \epsilon_{i,t} \quad (67)$$

on observations from $t = 0, \dots, \max\{E_i\} - 1$ and $E_i \neq 0$. We need to drop time period beyond $\max\{E_i\}$ because DID estimators for $CATT_{e,\ell}$ for $e + \ell \geq \max\{E_i\}$ do not exist as explained in the main text. We need to exclude cohort 0 from estimation because DID estimators for $CATT_{0,\ell}$ do not exist as explained in the main text. Note that among regressors we exclude interactions with $\mathbf{1}\{E_i = \max\{E_i\}\}$ and we exclude interactions with $D_{i,t}^{-1}$. Here $\mathbf{B}_{i,t}$ is a column vector collecting the interactions $\mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell$. Similarly, $\boldsymbol{\delta}$ is a column vector collecting the coefficients $\delta_{e,\ell}$ on $\mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell$. The matrix notation is used later to derive the asymptotic variance of IW estimators.

Step 2 Estimate the weights, which are cohort shares among cohorts that experience at least ℓ periods of treatment relative to the initial treatment.

Denote by $N_e := \sum_{i=1}^N \mathbf{1}\{E_i = e\}$ the number of units in cohort e . Denote by $h^\ell = \{e : 1 - \ell \leq e \leq \max\{E_i\} - 1 - \ell\}$ to be the set of cohorts that experience at least ℓ periods of treatment relative to the initial treatment. Below $\text{vec}(A)$ vectorizes matrix A by stacking its columns.

Define $\widehat{\mathbf{f}}^\ell$ to be a matrix with its $(t, e)^{th}$ entry equal to $\mathbf{1}\{t - e = \ell\} \cdot N_e / \sum_{e \in h^\ell} N_e$. Here $\mathbf{1}\{t - e = \ell\}$ indicates when cohort e experiences exactly ℓ periods of treatment and $N_e / \sum_{e \in h^\ell} N_e$ is equal to the sample share of units in cohort e among units that experience at least l periods of treatment. Denote by $\mathbf{f}^\ell$ the probability limit of $\widehat{\mathbf{f}}^\ell$, which is a matrix with its $(t, e)^{th}$ entry equal to $\mathbf{1}\{t - e = \ell\} \cdot \Pr\{E_i = t - l \mid E_i \in h^\ell\}$. For example, with $E_i \in \{0, 1, 2, 3\}$ for $T = 3$ and $\ell = 0$, we have $h^0 = \{1, 2\}$ and thus

$$\widehat{\mathbf{f}}^0 = \begin{pmatrix} \frac{N_1}{N_1 + N_2} & 0 \\ 0 & \frac{N_2}{N_1 + N_2} \end{pmatrix} \quad (68)$$

and its probability limit is

$$\mathbf{f}^0 = \begin{pmatrix} \Pr\{E_i = 1 \mid 1 \leq E_i \leq 2\} & 0 \\ 0 & \Pr\{E_i = 2 \mid 1 \leq E_i \leq 2\} \end{pmatrix}. \quad (69)$$

In proof below, we show that the weight matrix estimator $\widehat{\mathbf{f}}^\ell$ is asymptotically normal $\sqrt{N}(\text{vec}(\widehat{\mathbf{f}}^\ell) - \text{vec}(\mathbf{f}^\ell)) \rightarrow_d N(0, \Sigma_{f^\ell})$.

Step 3 Compute the IW estimator as the weighted sum of $\widehat{\delta}_{e,\ell}$ (estimated in Step 1) using weights (estimated

in Step 2).

To form an estimator alternative to the dynamic FE estimator $\hat{\mu}_g$ from Regression (4), we can use

$$\hat{v}_g := \frac{1}{|g|} \sum_{\ell \in g} \sum_{e \in h^\ell} \frac{N_e}{\sum_{e \in h^\ell} N_e} \hat{\delta}_{e,\ell} = \frac{1}{|g|} \sum_{\ell \in g} \text{vec}(\hat{\mathbf{f}}^\ell)^\top \hat{\boldsymbol{\delta}}. \quad (70)$$

With a few standard assumptions (which we present together below as Assumption 4) on Regression (26), we can show that the IW estimators are asymptotically normal.

Assumption 4. (The saturated regression assumptions).

1. There are observations from at least two cohorts that are not treated in $t = 0$.
2. Independent, identically distributed cross-sectional observations: $\{(E_i, \mathbf{Y}_i) : i = 1, 2, \dots, N\}$ are i.i.d. draws from their joint distribution where $\mathbf{Y}_i = (Y_{i,0}, \dots, Y_{i,T})^\top$ is a $T \times 1$ vector.
3. Large outliers are unlikely: $(\mathbf{B}_{i,t}, \epsilon_{i,t})$ have nonzero finite fourth moments.
4. Denote by $\ddot{\mathbf{B}}$ the data matrix, whose rows consist of $\ddot{\mathbf{B}}_{i,t}^\top$, double-demeaned version of $\mathbf{B}_{i,t}^\top$. Assume $\ddot{\mathbf{B}}$ has full rank. If $\ddot{\mathbf{B}}$ is reduced-rank because cohort e is empty, then discard regressors involving $\mathbf{1}\{E_i = e\}$.

Denote by $\boldsymbol{\delta}$ the probability limit of $\hat{\boldsymbol{\delta}}$, which is a vector of $CATT_{e,\ell}$. We next state the asymptotic distribution of the IW estimators specifically for v_ℓ , the average effect at relative time ℓ . Results for the more general case of v_g , which is the average effects across relative times $\ell \in g$, can be derived similarly. Note that we use a clustered variance-covariance structure to allow the possibility that $Y_{i,t}$ are dependent across t due to serial correlation.

Proposition 6. (Consistency and asymptotic normality of the IW estimators for v_ℓ). *Under the assumptions of Proposition 5 and Assumption 4, the IW estimator converges in probability to*

$$\hat{v}_\ell \rightarrow_p \sum_{e \in h^\ell} \Pr\{E_i = e \mid E_i \in h^\ell\} CATT_{e,\ell} = \text{vec}(\mathbf{f}^\ell)^\top \boldsymbol{\delta}. \quad (71)$$

The asymptotic distribution of this estimator is

$$\sqrt{N} \left(\hat{v}_\ell - \text{vec}(\mathbf{f}^\ell)^\top \boldsymbol{\delta} \right) \rightarrow_d N \left(0, \boldsymbol{\delta}^\top \Sigma_{f^\ell} \boldsymbol{\delta} + \Sigma_\ell \right) \quad (72)$$

for Σ_{f^ℓ} the asymptotic variance of $\sqrt{N}(\text{vec}(\hat{\mathbf{f}}^\ell) - \text{vec}(\mathbf{f}^\ell))$ where $\hat{\mathbf{f}}^\ell$ is the weight matrix estimator and

$$\mathbf{V}_{\ddot{\mathbf{B}}} = \sum_{t=0}^{\max\{E_i\}-1} E[\ddot{\mathbf{B}}_{i,t} \ddot{\mathbf{B}}_{i,t}^\top] \quad \Sigma_\ell = \text{vec}(\mathbf{f}^\ell)^\top \mathbf{V}_{\ddot{\mathbf{B}}}^{-1} \text{Var} \left(\sum_{t=0}^{\max\{E_i\}-1} \ddot{\mathbf{B}}_{i,t} \epsilon_{i,t} \right) \mathbf{V}_{\ddot{\mathbf{B}}}^{-1} \text{vec}(\mathbf{f}^\ell). \quad (73)$$